Question 1:

Install Apache or Nginx on your Linux server and verify the installation by accessing the default web page via your browser.

Answer:

- **Step 1: Install and start the web server (using Apache as standard):**
    - Update repository: sudo apt update
    - Install Apache: sudo apt install apache2 -y
    - Start and enable service: sudo systemctl start apache2 && sudo systemctl enable apache2
- **Step 2: Verification:**
    - Open a web browser on the local machine and navigate to http://localhost or http://127.0.0.1.
    - The browser renders the default "Apache2 Ubuntu Default Page," confirming the server daemon is actively running and serving HTTP traffic on port 80.

Question 2:

Create a folder named myflipkart in your web server's root directory and place an index.html file inside it that displays 'Welcome to MyFlipkart Deals!'.

Answer:

- **Step 1: Create the directory in the web root:**

sudo mkdir -p /var/www/html/myflipkart

- **Step 2: Create and populate the index.html file:**

Run the following command to generate the HTML file:

Bash

sudo bash -c 'cat <<EOF > /var/www/html/myflipkart/index.html

<!DOCTYPE html>

<html lang="en">

<head>

```
  <meta charset="UTF-8">
  <title>MyFlipkart Deals</title>
</head>
<body>
  <h1>Welcome to MyFlipkart Deals!</h1>
</body>
</html>
EOF'
```

- **Step 3: Adjust permissions so the web server can read the files:**

```
sudo chown -R www-data:www-data /var/www/html/myflipkart
sudo chmod -R 755 /var/www/html/myflipkart
```

Question 3:

Update your web server configuration so that visiting http://localhost/myflipkart shows your custom index.html page instead of the default server page.

Hint: For Apache, use the Alias directive; for Nginx, use a new location block.

Answer:

- **For Apache:**

    1. Open the default site configuration file:

```
sudo nano /etc/apache2/sites-available/000-default.conf
```

    2. Add the Alias directive and directory permissions inside the <VirtualHost *:80> block:

```
Apache
Alias /myflipkart /var/www/html/myflipkart

<Directory /var/www/html/myflipkart>
  Options Indexes FollowSymLinks
  AllowOverride None
```

```
    Require all granted
</Directory>
```

3. Test configuration and restart Apache:

```
sudo apache2ctl configtest
sudo systemctl restart apache2
```

- **For Nginx (Alternative):**

1. Add a location block inside /etc/nginx/sites-available/default:

```
Nginx
location /myflipkart {
    alias /var/www/html/myflipkart;
    index index.html;
}
```

2. Test and reload: sudo nginx -t && sudo systemctl reload nginx

- **Verification:**

Open http://localhost/myflipkart in a browser. It displays the custom heading "Welcome to MyFlipkart Deals!".

Question 4:

Add a favicon to your static site by placing a favicon.ico file in the correct location and updating your HTML to reference it.

Constraint: Use any small image and rename it to favicon.ico for this exercise.

Answer:

- **Step 1: Place a favicon into the site directory:**

Copy any available icon or image to the website folder:

```
sudo cp /usr/share/icons/hicolor/16x16/apps/*.png /var/www/html/myflipkart/favicon.ico
```

(Or download/create one directly inside /var/www/html/myflipkart/favicon.ico)

- **Step 2: Ensure proper file ownership and permissions:**

```
sudo chown www-data:www-data /var/www/html/myflipkart/favicon.ico
```

sudo chmod 644 /var/www/html/myflipkart/favicon.ico

- **Step 3: Update index.html to reference the favicon:**

Edit /var/www/html/myflipkart/index.html to include the <link> tag within the <head> block:

HTML

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>MyFlipkart Deals</title>
  <link rel="icon" type="image/x-icon" href="/myflipkart/favicon.ico">
</head>
<body>
  <h1>Welcome to MyFlipkart Deals!</h1>
</body>
</html>
```

- **Verification:**

Hard-refresh http://localhost/myflipkart (using Ctrl + F5). The selected icon appears in the browser tab beside the page title.